# Multiclass Classification

## Designing Codebooks and ECOC

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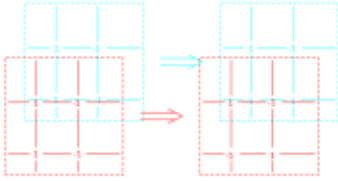

### Learning goals

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- Know what a codebook is
- Understand that codebooks generalize one-vs-one and one-vs-rest
- Know how to define a good codebook and error-correcting output codes (ECOC)
- Know how randomized hill-climbing algorithm is used to find good codebooks

# CODEBOOKS

- We have already seen that we can write down principles like one-vs-rest and one-vs-one reduction compactly by so-called **codebooks**.
- During training, a scoring classifier is trained for each column.
- The k-th row is called **codeword** for class $k$.
- Knowing the principle of **codebooks**, we can define multiclass-to-binary reductions quite flexibly.
- We can now ask ourselves, how to create optimal codebooks.

| Class | $f_1(\mathbf{x})$ | $f_2(\mathbf{x})$ | $f_3(\mathbf{x})$ |
|---|---|---|---|
| 1 | 1 | -1 | -1 |
| 2 | -1 | 1 | -1 |
| 3 | - 1 | -1 | 1 |

| Class | $f_1(\mathbf{x})$ | $f_2(\mathbf{x})$ | $f_3(\mathbf{x})$ |
|---|---|---|---|
| 1 | 1 | -1 | 0 |
| 2 | -1 | 0 | 1 |
| 3 | 0 | 1 | -1 |

Left: one-vs-rest codebook. Right: one-vs-one codebook.

# CODEBOOKS: DECIDING LABELS

For a general codebook, once we trained the classifiers, how to predict the class $\hat{y}$ for a new input $\mathbf{x}$?

- When a new sample $\mathbf{x}$ is going to be classified, all classifiers $f_k$ are applied to $\mathbf{x}$, scores are potentially transformed and turned into binary labels by $\text{sgn}\left(f_k(\mathbf{x})\right)$.

| Class | $f_1(\mathbf{x})$ | $f_2(\mathbf{x})$ | $f_3(\mathbf{x})$ |
|---|---|---|---|
| 1 | 1 | 1 | 0 |
| 2 | -1 | 1 | 1 |
| 3 | 0 | -1 | -1 |
| $\text{sgn}(\vec{f}(\mathbf{x}))$ | **-1** | **1** | **- 1** |

- We obtain a code for the observation $\mathbf{x}$ for which we can calculate the distance to the codewords of the other classes. This can be done by **Hamming distance** (counting the number of bits that differ) or by $L1$-distance.

# CODEBOOKS: DECIDING LABELS / 2

- For example, the $L1$-distance between $\text{sgn}\left(\hat{f}(\mathbf{x})\right) = (-1, 1, -1)$ and the class 1 codeword $(1, 1, 0)$ is 3.
- We can do so for all the classes to obtain respective distances:

| **Classes** | Dist |
|---|---|
| 1 | 3 |
| 2 | 2 |
| 3 | 3 |

The distance for class 2 is minimal, therefore we predict class 2 for the input $\mathbf{x}$.

# ERROR-CORRECTING CODES (ECOC) / 2

Another desirable property is **column separation**:

- Columns should be uncorrelated.
- If two columns $k$ and $l$ are similar or identical, a learning algorithm will make similar (correlated) mistakes in learning $f_k$ and $f_l$.
- Error-correcting codes only succeed if the errors made in the individual classifiers are relatively uncorrelated, so that the number of simultaneous errors in many classifiers is small.
- Errors in classifiers $f_k$ and $f_l$ will also be highly correlated if the bits in those columns are complementary.
- Try to ensure that columns are neither identical nor complementary.

$\rightarrow$ **We want to maximize distances between rows, and want the distances between columns to not be too small (identical columns) or too high (complementary columns).**

# ERROR-CORRECTING CODES (ECOC) / 3

**Remark:**

- In general, if there are $k$ classes, there will be at most $2^{k-1} - 1$ usable binary columns.
- For example for $k = 3$, there are only $2^3 = 8$ possible columns. Of these, half are complements of the other half. The columns that only contain 1s or the one that only contains $-1s$ are also not usable.

| Class | $f_1(\mathbf{x})$ | $f_2(\mathbf{x})$ | $f_3(\mathbf{x})$ | $f_4(\mathbf{x})$ | $f_5(\mathbf{x})$ | $f_6(\mathbf{x})$ | $f_7(\mathbf{x})$ | $f_8(\mathbf{x})$ |
|---|---|---|---|---|---|---|---|---|
| 1 | -1 | -1 | -1 | -1 | 1 | 1 | 1 | 1 |
| 2 | -1 | - 1 | 1 | 1 | -1 | -1 | 1 | 1 |
| 3 | -1 | 1 | -1 | 1 | -1 | 1 | -1 | 1 |

# ERROR-CORRECTING CODES (ECOC) / 4

Assume we have the budget to train $L$ binary classifiers and now want to find an error-correcting code with maximal row and column separation.

- For only few classes $g \leq 11$, exhaustive search can be performed and a codebook that has good row and column separation is chosen.
- However, for many classes $g > 11$, it becomes more and more challenging to find the optimal codebook with codewords of length $L$.
- *Dietterich et al.* employed a randomized hill-climbing algorithm for this task.